

COMP 1101 - Midterm 2

Olcay Taner YILDIZ, Farshad MIRAMIRKHANI, Onder UMUT, Bora AKSAR

1. What is the value of num after the following loop is executed?

```
for num = 1 : 5
end
disp(num);
```

- a) 2
- b) 3
- c) 4
- d) 5
- e) 6

2. Which of the following codes produce the following output?

```
Hello
World
Hello
World
Hello
World
```

a)

```
for i = 1 : 6
    disp("Hello");
    disp("World");
end
```

b)

```
for i = 1 : 3
    disp("Hello ");
    disp("World");
end
```

c)

```
for i = 1 : 2 : 6
    disp("Hello ");
end
for i = 2 : 2 : 6
    disp("Hello ");
end
```

d)

```
for i = 1 : 3
    disp("Hello ");
end
for i = 1 : 3
    disp("World");
end
```

e)

```
for i = 1 : 6
    disp("Hello");
end
for i = 1 : 6
    disp("World");
end
```

3. Which of the following codes print numbers from 23 to 50 in one line?

a)

```
displayed = "";
for i = 23 : 49
    displayed = displayed + "i ";
end
disp(displayed);
```

b)

```
for i = 23 : 49
    disp(i);
end
```

c)

```
displayed = "";
for i = 23 : 50
    displayed = displayed + "i ";
end
disp(displayed);
```

d)

```
displayed = "";
for i = 23 : 50
    displayed = displayed + i;
end
disp(displayed);
```

e)

```
for i = 23 : 50
    disp(i);
end
```

4. Which of the following codes produce the following output?

```
***
```

```
***
```

```
***
```

a)

```
for i = 1 : 3
    disp("*");
```

```
disp("*");
end
```

b)

```
for i = 1 : 3
    disp("*");
    disp("*");
    disp("*");
end
```

c)

```
for i = 1 : 2 : 6
    disp("*");
end
for i = 2 : 2 : 6
    disp("*");
end
```

d)

```
for i = 1 : 3
    displayed = "*";
    displayed = displayed + "*";
    displayed = displayed + "*";
    disp(displayed);
end
```

e)

```
for i = 1 : 3
    displayed = "*";
    displayed = displayed + "*";
    disp(displayed);
end
```

5. What is the output of the following code?

```
for x = 1 : 4
    disp(x);
    if x == 3
        x = x + 1;
    else
        x = x + 2;
    end
end
```

a)

```
1
4
```

b)

```
1
2
3
4
```

c)

```
1
2
3
4
5
```

d)

```
1
3
5
```

e)

```
1
3
4
```

6. What is missing in the following code fragment?

```
while count <= 8
    disp("Hi");
    count = count + 1;
end
```

- a) Setting the initial value of the loop variable
- b) Testing the value of the loop variable
- c) Increasing the value of the loop variable
- d) Decreasing the value of the loop variable
- e) None of the above

7. Let say you want to write a program which will take a positive number N as input from the user and prints the sum of integers until N . The following statements are given.

```
sum = sum + i;
disp(sum);
for i = 1 : N
    int sum = 0;
    N = input("Enter number");
end
```

What is the correct order of statements if you want to accomplish this task?

- a) 4, 2, 3, 5, 1, 6
- b) 5, 2, 3, 1, 6, 4
- c) 4, 5, 3, 1, 6, 2
- d) 4, 3, 5, 1, 2, 6
- e) 2, 4, 5, 1, 6, 3

8. Which of the following code blocks is the same as the following code block?

```
for i = 6 : -2 : 1
    disp("2x" + i + "=" + i);
end
```

a)

```
disp("2x6=6");
disp("2x4=4");
disp("2x2=2");
```

b)

```
disp("2x6=12");
disp("2x4=8");
disp("2x2=4");
```

c)

```
disp("2x6=12");
disp("2x5=10");
disp("2x4=8");
disp("2x3=6");
disp("2x2=4");
disp("2x1=2");
```

d)

```
disp("2x6=6");
disp("2x5=5");
disp("2x4=4");
disp("2x3=3");
disp("2x2=2");
disp("2x1=1");
```

e)

```
disp("2x3=6");
disp("2x2=4");
disp("2x1=2");
```

9. Consider the following for loop

```
for x = A: B : C
end
```

How do you convert this for loop into a while loop?

a)

```
x = A;
while x <= C
end
x = x + B;
```

b)

```
x = B;
while x <= A
    x = x + C
end
```

c)

```
x = A;
while x <= C
    x = x + B;
end
```

d)

```
x = A;
x = x + B;
while x <= C
end
```

e)

```
x = A;
while x <= B
    x = x + C;
end
```

10. Which of the following code fragments calculate the following sum:

$$\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{N}$$

a)

```
total = 0;
i = 1;
while i >= N
    total = total + 1 / i;
    i = i + 1;
end
```

b)

```
total = 0;
for i = 2 : N
    total = total + 1 / i;
end
```

c)

```
total = 0;
i = 1;
while i <= N
    total = total + 1.0 / i;
end
```

d)

```
total = 0;
for i = 1 : N
    total = total + 1 / i;
end
```

e)

```
total = 0;
i = 1;
while i <= N
    i = i + 1;
    total = total + 1.0 / i;
end
```

11. What is the output of the following program?

```
x = 5;
while x < 10
    x = x + 1;
    if x > 7
        break;
    end
    disp(x);
end
```

a)

```
6
7
8
```

b)

```
5
6
7
```

c)

```
5
6
7
8
9
```

d)

6
7

e)

6
7
8
9

12. What is the output of the following code?

```
total = 0;
i = 0;
while i <= 5
    total = total + i;
    i = i + 1;
end
disp(i + " " + total);
```

- a) 5 10
- b) 6 10
- c) 4 10
- d) 5 15
- e) 6 15

13. What will be the output of the following program?

```
for i = 1 : 3
    displayed = "";
    for j = 1 : 3
        displayed = displayed + "*";
    end
    disp(displayed);
end
```

- a) ***

- b) ***
**
*
- c) *
**

- d) *****
- e) ***
*

14. What will be the output of the following program?

```
total = 0;
for i = 1 : 5
    for j = 1 : 5
```

```
        for k = j : 5
            total = total + 1;
        end
    end
end
disp(total);
```

- a) 55
- b) 35
- c) 210
- d) 75
- e) 110

15. What will be the output of the following program?

```
total = 0;
for i = 1 : 10
    for j = i : 10 - i
        total = total + 1;
    end
end
disp(total);
```

- a) 55
- b) 110
- c) 15
- d) 25
- e) 10

16. Suppose you are given a method named **gcd** that calculates and returns the greatest common divisor of two integer numbers. Which of the following is the correct statement to call **gcd** method?

- a) result = **gcd**(int x, int y)
- b) result = **gcd**(x, y)
- c) **gcd**(x, y)
- d) **gcd**(int x, int y)
- e) result = call **gcd**(x, y)

17. Suppose you are given method named **maxDigit** that takes an integer number as input parameter and returns its maximum digit. For example, for input 574186 your method should return 8. Which of the following is a correct method header for **maxDigit** method?

- a) **function** maxDigit(n)
- b) **function** result = maxDigit(n)
- c) maxDigit(n)
- d) result = maxDigit(n)
- e) maxDigit

18. Consider the following program:

```
N = input("Enter number");
k = 1;
```

```

for i = 1 : N
    k = k * i;
end
disp(k);

```

How do you convert this program into a method?

a)

```

function result = f(N)
    k = 1;
    for i = 1 : N
        k = k * i;
    end
    disp(k);
end

```

b)

```

function result = f(N)
    k = 1;
    for i = 1 : N
        k = k * i;
    end
    result = k;
end

```

c)

```

function f()
    k = 1;
    for i = 1 : N
        k = k * i;
    end
    result = k;
end

```

d)

```

function result = f(N)
    N = input("Enter number")
    k = 1;
    for i = 1 : N
        k = k * i;
    end
    result = k;
end

```

e)

```

function f()
    N = input("Enter number")
    k = 1;
    for i = 1 : N
        k = k * i;
    end
    disp(k);
end

```

19. Let say you have the following function defined:

```

function printStar(n)
    displayed = ""
    for i = 1 : n
        displayed = displayed + "*";
    end
end

```

```

disp(displayed);
end

```

Which of the following code fragments produce the following output?

```

*
**
***
**
*

```

a)

```

printStar (1);
printStar (2);
printStar (3);
printStar (2);
printStar (1);

```

b)

```

printStar (1);
printStar (2);
printStar (3);

```

c)

```

printStar (3);
printStar (2);
printStar (1);

```

d)

```

printStar (3);
printStar (2);
printStar (1);
printStar (2);
printStar (3);

```

e)

```

printStar (3, 3);

```

20. Let say you have the function isPrime

```

function result = isPrime(N)

```

which returns **true** if N is prime, **false** otherwise. Which of the following code fragments prints all prime numbers less than 100?

a)

```

disp(isPrime(100));

```

b)

```

for i = 2 : 100
    disp(isPrime(i));
end

```

c)

```

for i = 2 : 100
    if isPrime(i) == true
        disp(i);
    end
end

```

d)

```
for i = 2 : 100
    disp(i);
end
```

e)

```
isPrime(100);
```

21. What will be the output when we call the following function with the expression nPrint(" Ali ", 4);

```
function nPrint(message, n)
    i = 1;
    while i <= n
        disp(message);
        i = i + 2;
    end
end
```

- a) AliAli
- b) AliAliAli
- c) Ali
Ali
Ali
Ali
- d) Ali
Ali
- e) AliAliAliAli

22. Let say you have the functions factorial and power

```
function result = factorial (N)
function result = power(x, y)
```

which calculate $N!$ and x^y respectively. Which of the following code fragments then calculate the following sum:

$$\frac{1!}{2^0} + \frac{2!}{2^1} + \dots + \frac{(N+1)!}{2^N}$$

a)

```
sum = factorial(N+1) / power(2, N);
```

b)

```
total = 1;
for i = 1 : N
    total = total + ( factorial (i) + 0.0) / power(2, N);
end
```

c)

```
total = 0;
for i = 1 : N
    total = total + ( factorial (N+1) + 0.0) / power(2, i);
end
```

d)

```
total = 0;
for i = 0 : N
    total = total + factorial (i+1) / power(2, i);
end
```

e)

```
total = 0;
for i = 1 : N
    total = total + ( factorial (i) + 0.0) / power(2, i+1);
end
```

23. Let say you have the function isEven

```
function result = isEven(N)
```

which returns **true** if N is even, **false** otherwise. Which of the following code fragments prints all positive even numbers less than 10?

a)

```
disp(isEven(10));
```

b)

```
for i = 2 : 9
    disp(isEven(i));
end
```

c)

```
for i = 2 : 9
    if isEven(i) == true
        disp(i);
    end
end
```

d)

```
for i = 2 : 9
    disp(i);
end
```

e)

```
isEven(10);
```

24. Which of the following is the correct prototype of a method?

- 1) function a

- 2) function a()
- 3) a()
- 4) function result = a(N)
- a) 1, 2
- b) 1
- c) 1, 2, 3
- d) 2, 4
- e) 1, 2, 3, 4

25. Let say you have the following function defined:

```
function printStar(n)
    displayed = ""
    for i = 1 : n
        displayed = displayed + "*";
    end
    disp(displayed);
end
```

Which of the following code fragments produce the following output?

```
*
**
*
**
*
**
```

a)

```
for i = 0 : 2
    printStar (3);
end
```

b)

```
for i = 0 : 6
    printStar (1);
    printStar (2);
end
```

c)

```
for i = 1 : 1
    printStar (1);
end
for i = 1 : 1
    printStar (2);
end
```

d)

```
for i = 0 : 2
    printStar (2);
end
```

e)

```
for i = 1 : 3
    printStar (3);
end
```